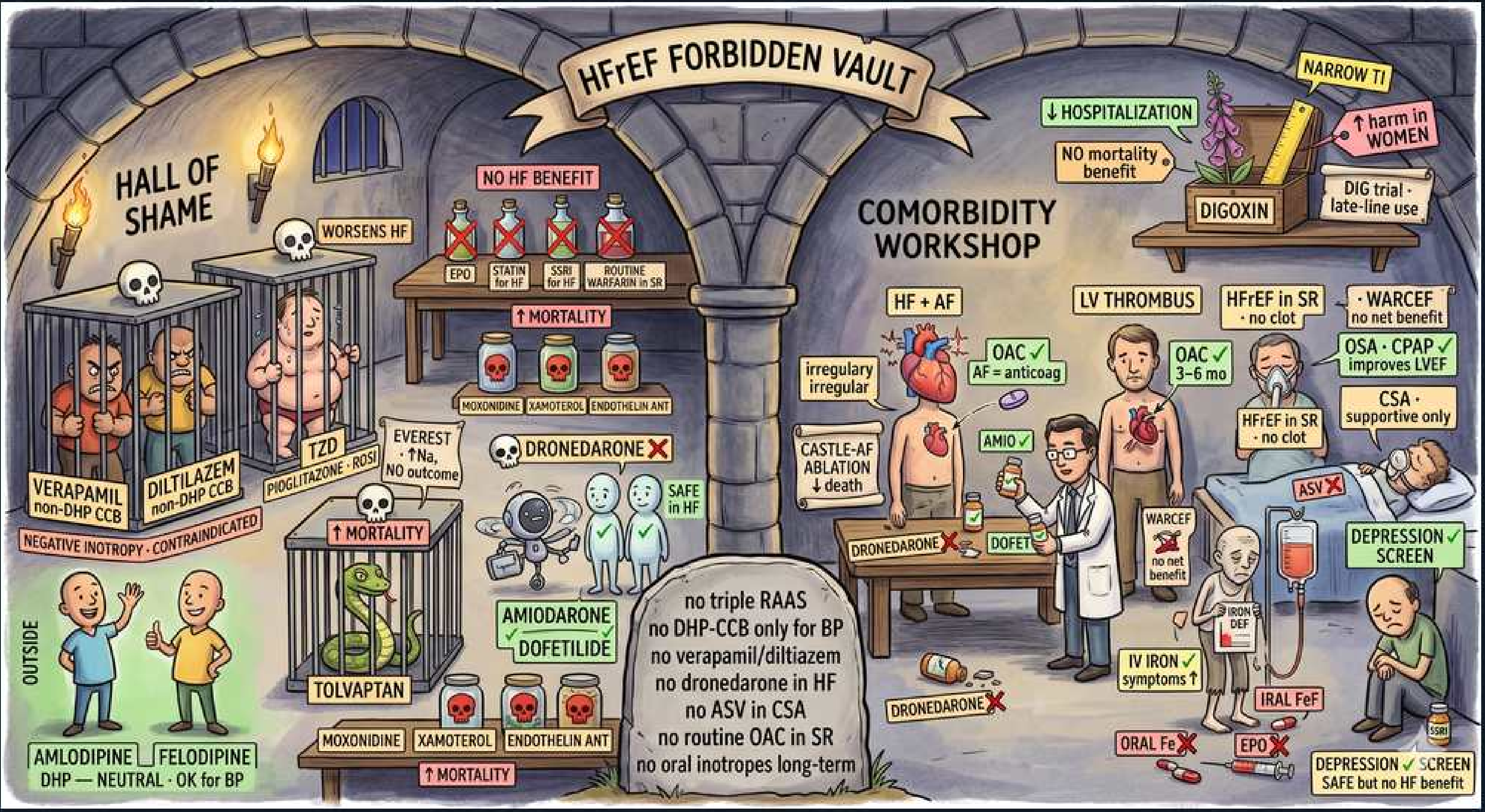
Heart Failure — SCENE 6: HF Hall of Shame & Comorbidities



1. Verapamil — non-DHP CCB

WHAT YOU SEE

A prisoner locked behind bars in the HALL OF SHAME dungeon, labeled VERAPAMIL with a skull and 'NEGATIVE INOTROPY / CONTRAINDICATED' — the cage = banned in HFrEF because it drains the already-weak pump.

WHAT IT ENCODES

Non-dihydropyridine CCBs — verapamil (the most negatively inotropic) and diltiazem — are CONTRAINDICATED in HFrEF because their negative inotropy depresses an already failing ventricle and can precipitate decompensation (and bradycardia/AV block, especially combined with a beta-blocker). They are acceptable only for rate control in HFpEF (preserved EF) where contractility is intact.

BOARD TRAP

Trap: an AF + HFrEF (EF 30%) patient needing rate control, answer 'start verapamil/diltiazem.' WRONG — non-DHP CCBs are contraindicated in HFrEF. Discriminator: use a beta-blocker for rate control (and digoxin if needed); reserve verapamil/diltiazem for AF with PRESERVED EF. Combining a non-DHP CCB with a beta-blocker also risks profound bradycardia/heart block.

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2. Diltiazem caged

WHAT YOU SEE

A second prisoner caged right beside verapamil, labeled DILTIAZEM (non-DHP CCB) — same cage, same crime: a negative inotrope locked away from the reduced-EF heart.

WHAT IT ENCODES

Diltiazem, like verapamil, is a negative-inotrope non-DHP CCB that worsens HFrEF and should be avoided for AF rate control when LVEF is reduced. It is somewhat less negatively inotropic than verapamil but still unsafe in low-EF HF; both also slow AV conduction.

BOARD TRAP

Trap: 'diltiazem is the safe non-DHP CCB I can use in low-EF heart failure because it's gentler than verapamil.' WRONG — both verapamil AND diltiazem are contraindicated in HFrEF. Discriminator: 'less negatively inotropic' does not equal 'safe' in reduced EF; use a beta-blocker for rate control instead.

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3. TZD (pioglitazone)

WHAT YOU SEE

A pill bottle labeled TZD with a 'WORSENS HF' arrow/skull — the glitazones are jailed in the shame hall because they make the body retain fluid and flood the failing heart.

WHAT IT ENCODES

Thiazolidinediones (pioglitazone, rosiglitazone) cause renal sodium and fluid RETENTION via ENaC upregulation, leading to edema and increased HF hospitalization; they are contraindicated in NYHA III–IV HF and should be avoided in HFrEF generally. (Pioglitazone also carries bladder-cancer and fracture cautions.)

BOARD TRAP

Trap: 'TZDs are weight-neutral/cardiac-neutral, so pioglitazone is fine for this diabetic with HFrEF.' WRONG — TZDs cause fluid retention and worsen HF (contraindicated in NYHA III–IV). Discriminator: in a diabetic with HF, choose an SGLT2i (which BENEFITS HF) or metformin; avoid TZDs and the DPP-4 inhibitor saxagliptin (and alogliptin), which increased HF hospitalization (SAVOR-TIMI).

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4. NSAIDs

WHAT YOU SEE

A scatter of NSAID pills in the 'worsens HF' column with a skull — the painkillers that retain salt/water and blunt diuretics, tipping HF into decompensation (the 'triple whammy').

WHAT IT ENCODES

NSAIDs (and COX-2 inhibitors) inhibit renal prostaglandins → sodium/water retention, vasoconstriction with rising afterload, blunting of diuretic and ACEi/ARB efficacy, and worsening renal function; they are a classic precipitant of acute HF decompensation. Combining NSAIDs with an ACEi/ARB + diuretic is the nephrotoxic 'triple whammy.'

BOARD TRAP

Trap: a HF patient with arthritis pain is given an NSAID for symptom control. WRONG — NSAIDs precipitate HF decompensation and AKI. Discriminator: prefer acetaminophen (or topical agents) for analgesia in HF; the 'triple whammy' (NSAID + ACEi/ARB + diuretic) is a high-yield cause of acute kidney injury and fluid overload.

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5. Amlodipine / Felodipine — neutral

WHAT YOU SEE

AMLODIPINE and FELODIPINE standing free OUTSIDE the dungeon, tagged 'DHP — NEUTRAL, OK for BP' — the DHP CCBs that are safe (just no mortality benefit), unlike their caged non-DHP cousins.

WHAT IT ENCODES

Amlodipine and felodipine are dihydropyridine CCBs that are HEMODYNAMICALLY NEUTRAL in HFrEF (PRAISE, V-HeFT III) — they do not improve survival but are SAFE add-ons for refractory hypertension or angina when GDMT alone is insufficient. They lack significant negative inotropy, unlike the non-DHP CCBs.

BOARD TRAP

Trap: 'all calcium channel blockers are contraindicated in HFrEF.' WRONG — the DHP CCBs amlodipine and felodipine are safe (neutral) and can be used for BP/angina. Discriminator: the contraindication applies to the NON-DHP CCBs (verapamil, diltiazem); amlodipine/felodipine are the acceptable CCBs in reduced EF (just no mortality benefit).

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6. Tolvaptan — no mortality benefit

WHAT YOU SEE

A caged figure labeled TOLVAPTAN with 'EVEREST / no outcome' — the aquaretic locked up because it raises sodium and eases symptoms but buys NO survival.

WHAT IT ENCODES

Tolvaptan is a vasopressin V2-receptor antagonist (aquaretic) that raises serum sodium and relieves congestion/dyspnea short-term but provides NO mortality or long-term morbidity benefit in HF (EVEREST). Its niche is hypervolemic/euvolemic HYPONATREMIA; watch for over-rapid sodium correction (osmotic demyelination) and hepatotoxicity.

BOARD TRAP

Trap: 'add tolvaptan to reduce mortality / as routine therapy in decompensated HF.' WRONG — EVEREST showed symptomatic/sodium improvement but no survival benefit. Discriminator: tolvaptan is for symptomatic hyponatremia, not a survival drug; correct Na+ slowly to avoid osmotic demyelination.

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7. Dronedarone — increases mortality

WHAT YOU SEE

DRONEDARONE drawn with a skull and a big '↑MORTALITY' arrow — the antiarrhythmic that kills in HF, marked as forbidden, not safe.

WHAT IT ENCODES

Dronedarone is CONTRAINDICATED in symptomatic HF (NYHA III–IV or recently decompensated) and in permanent AF — it INCREASED mortality in decompensated HFrEF (ANDROMEDA) and increased CV events in permanent AF (PALLAS). It also causes hepatotoxicity.

BOARD TRAP

Trap: 'dronedarone is the preferred/safe antiarrhythmic in heart failure because it lacks amiodarone's toxicities.' WRONG — dronedarone increases mortality in HF and is contraindicated. Discriminator: in HFrEF the antiarrhythmics that are mortality-NEUTRAL and safe are amiodarone and dofetilide; dronedarone is forbidden.

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8. Amiodarone / dofetilide — safe

WHAT YOU SEE

AMIODARONE and DOFETILIDE standing OUTSIDE the cages with a green 'SAFE / mortality-neutral' tag — the only two antiarrhythmics allowed to walk free in HFrEF.

WHAT IT ENCODES

Amiodarone and dofetilide are the antiarrhythmics considered mortality-NEUTRAL and safe to use for AF/VT in HFrEF. Amiodarone has extensive organ toxicities (thyroid, pulmonary fibrosis, hepatic, corneal/skin) requiring monitoring; dofetilide requires in-hospital initiation with QT/renal dosing because of torsades risk. Class IC agents (flecainide, propafenone) are CONTRAINDICATED in structural heart disease/HF (CAST).

BOARD TRAP

Trap: using a class IC agent (flecainide/propafenone) for AF in a patient with HFrEF or prior MI. WRONG — class IC drugs are proarrhythmic and increase mortality in structural heart disease (CAST). Discriminator: in structural HF, restrict to amiodarone or dofetilide; flecainide/propafenone are only for structurally NORMAL hearts.

9. Digoxin — narrow TI

WHAT YOU SEE

DIGOXIN in the top corner flagged 'NARROW TI' and 'worse in WOMEN' — the narrow-therapeutic-index drug that cuts hospitalizations but not deaths and toxifies easily.

WHAT IT ENCODES

Digoxin reduces HF HOSPITALIZATION but has NO mortality benefit (DIG trial); reserve it for persistent symptoms despite GDMT or for AF rate control. It has a narrow therapeutic index — target a low level (0.5–0.9 ng/mL); toxicity (nausea, confusion, visual halos/xanthopsia, arrhythmias) is potentiated by HYPOKALEMIA, hypomagnesemia, renal impairment, and is worse in women and the elderly.

BOARD TRAP

Trap: 'digoxin lowers mortality in HFrEF.' WRONG — DIG showed reduced hospitalizations only, with a neutral effect on mortality (and possible harm at higher levels/in women). Discriminator: digoxin is a symptom/hospitalization and rate-control drug, not a survival drug; keep levels 0.5–0.9 ng/mL and correct K+/Mg to avoid toxicity.

10. LV thrombus → anticoagulate

WHAT YOU SEE

A heart drawn with a clot inside it labeled LV THROMBUS → 'OAC 3-6 mo' — a visible clot means start anticoagulation, not just aspirin.

WHAT IT ENCODES

A documented LV thrombus (common after large anterior MI with apical akinesis, or in severe HFrEF/LV aneurysm) warrants anticoagulation, typically with warfarin (most evidence) or a DOAC, for at least 3–6 months with follow-up imaging to confirm resolution. This reduces systemic/embolic stroke risk.

BOARD TRAP

Trap: a patient with a freshly diagnosed LV thrombus is managed with antiplatelet therapy alone. WRONG — an established LV thrombus requires therapeutic ANTICOAGULATION (3–6 months), not just aspirin. Discriminator: thrombus present → anticoagulate; this differs from the no-thrombus, sinus-rhythm HFrEF patient who should NOT be routinely anticoagulated.

11. HFrEF in sinus rhythm — no routine OAC

WHAT YOU SEE

An 'HFrEF in SR' panel showing a clean heart with NO clot and 'no OAC / WARCEF' — low EF in sinus rhythm with no clot is NOT a reason to anticoagulate.

WHAT IT ENCODES

HFrEF in SINUS RHYTHM without AF, LV thrombus, or a prior thromboembolic event should NOT be routinely anticoagulated — WARCEF showed warfarin vs aspirin had no net benefit (fewer ischemic strokes offset by more major bleeding). Reserve anticoagulation for a specific indication (AF, thrombus, mechanical valve, VTE).

BOARD TRAP

Trap: 'low EF causes stasis, so anticoagulate every HFrEF patient to prevent stroke.' WRONG — routine anticoagulation in sinus-rhythm HFrEF is not beneficial and increases bleeding (WARCEF). Discriminator: anticoagulate only with a discrete indication (AF, LV thrombus, mechanical valve, recent VTE), not for low EF alone.

12. AF + HF → anticoagulate

WHAT YOU SEE

An 'AF + HF' panel with irregular squiggly rhythm and 'OAC = anticoag' — atrial fibrillation plus HF (HF itself scores a point) means anticoagulate.

WHAT IT ENCODES

Atrial fibrillation in a HF patient warrants anticoagulation based on CHA2DS2-VASc — and HF (the 'C') itself contributes a point, so essentially all AF + HF patients qualify. DOACs are preferred over warfarin for nonvalvular AF (except mechanical valves/moderate-severe mitral stenosis). Manage rate (beta-blocker ± digoxin) and consider rhythm control where appropriate.

BOARD TRAP

Trap: deferring anticoagulation in an AF + HF patient because the CHA2DS2-VASc was 'borderline.' WRONG — HF scores a point, so AF + HF almost always meets the anticoagulation threshold. Discriminator: AF + HF → anticoagulate (DOAC preferred in nonvalvular AF); contrast with sinus-rhythm HFrEF where routine OAC is not indicated.

13. IV iron for iron deficiency

WHAT YOU SEE

An IV iron infusion bag labeled 'IRAL FeP' with oral iron crossed out (oral Fe X) — IV iron repletes the deficiency that oral iron can't fix in HF.

WHAT IT ENCODES

Iron deficiency (ferritin <100, or 100–299 with TSAT <20%) is common in HFrEF even WITHOUT anemia and worsens symptoms/exercise capacity. IV iron (ferric carboxymaltose — FAIR-HF, CONFIRM-HF, AFFIRM-AHF) improves symptoms, QOL, and reduces HF hospitalization; ORAL iron is ineffective in HF (IRONOUT-HF, due to poor absorption from hepcidin upregulation).

BOARD TRAP

Trap: treating documented iron deficiency in a HFrEF patient with oral ferrous sulfate. WRONG — oral iron does not improve outcomes in HF (IRONOUT). Discriminator: use INTRAVENOUS iron (ferric carboxymaltose); also note iron deficiency is worth treating even when hemoglobin is normal (non-anemic iron deficiency).

14. Depression — screen, SSRI no benefit

WHAT YOU SEE

A 'DEPRESSION SCREEN' panel noting SSRIs are 'SAFE but no HF benefit' — screen and treat the mood, but don't expect the pill to improve heart-failure outcomes.

WHAT IT ENCODES

Depression is common in HF and associated with worse outcomes, so SCREEN for it; however, SSRIs (e.g., sertraline in SADHART-CHF, escitalopram in MOOD-HF) are SAFE but did NOT improve depression scores or cardiovascular outcomes in HF trials. Avoid tricyclic antidepressants (anticholinergic/arrhythmogenic). Consider cognitive-behavioral therapy and exercise.

BOARD TRAP

Trap: 'start an SSRI to improve heart-failure outcomes/prognosis in a depressed HF patient.' WRONG — SSRIs are safe but show no mortality, hospitalization, or even reliable mood benefit in HF trials. Discriminator: screen and manage depression for its own sake, but do not expect (or choose) an SSRI to improve HF outcomes; avoid TCAs in HF.